Multivariable Taylor Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of multivariable taylor theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AC-03 / Advanced Calculus / Report v1

I. TARGET AND IMPORT CONTRACT

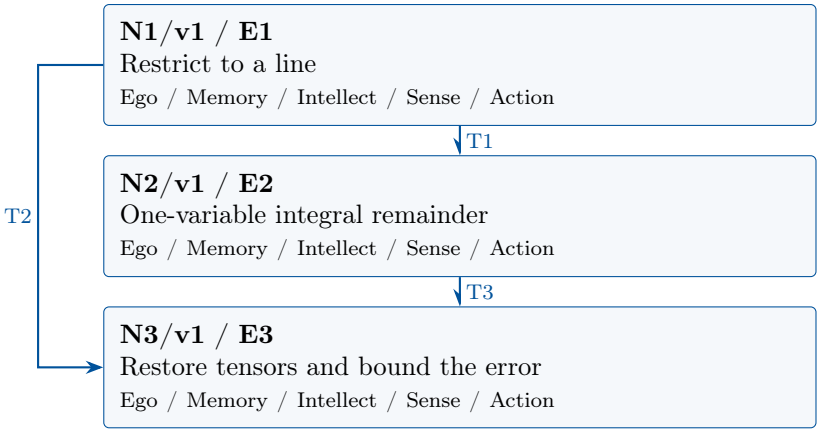
Theorem I.1. *Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ be C^{k+1}, with the segment from a to $a+h$ in U. Prove $f(a+h) = \sum_{j=0}^{k} D^j f(a)[h,\dots,h]/j! + \frac{1}{k!}\int_0^1 (1-t)^k D^{k+1} f(a+th)[h,\dots,h]\,dt$.*

A. Foundations and permitted imports

$D^j\ f(a)$ denotes the symmetric j-linear derivative; $D^0 f = f$. Import the chain rule and AC-02. The segment is contained in open U and f is $C^{(k+1)}$ near it, so $g(t) = f(a+th)$ has the necessary derivatives on a neighborhood of[0,1]. Derive the one-variable integral remainder below.

Here $k \geq 0$ is an integer, $U \subset \mathbb{R}^n$ is open, and the entire segment from a to $a+h$ lies in U.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The directional derivative identities along the entire segment.

Receiving activity. N2: One-variable integral remainder

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: $g \in C^{(k+1)}$ near[0,1]; prior exports E1/v1

T2: N1/E1/v1 to N3. The directional derivative identities along the entire segment.

Receiving activity. N3: Restore tensors and bound the error

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Directional identities and scalar formula; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. The exact one-variable Taylor formula with integral remainder.

Receiving activity. N3: Restore tensors and bound the error

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Directional identities and scalar formula; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Restrict to a line

Object and role. Reduce the multivariable increment to a scalar parameter.

Inputs. $f \in C^{(k+1)}$; $a + th \in U$ for $0 \le t \le 1$; foundational imports stated in §1

E — Ego

Own E1/v1: The directional derivative identities along the entire segment.

M — Memory

Retain $f \in C^{(k+1)}$; $a + th \in U$ for $0 \le t \le 1$; foundational imports stated in §1; keep their assumptions and versions attached.

I — Intellect

Reduce the multivariable increment to a scalar parameter.

S — Sense

Endpoint-only domain membership is insufficient if the segment leaves U.

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Define $g(t) = f(a + th)$.

N1.2 Repeated chain rule along the affine path yields $g^{(j)}(t) = D^j f(a + th)[h, \dots, h]$ for $0 \le j \le k + 1$.

N1.3 For $j = 0$ this means the function itself.

N1.4 Compactness of the segment gives a neighborhood on which these restrictions are defined; no global convexity of U is assumed.

Export E1/v1. The directional derivative identities along the entire segment.

Acceptance gate. Endpoint-only domain membership is insufficient if the segment leaves U. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Endpoint-only domain membership is insufficient if the segment leaves U. Recheck dependent nodes N2, N3.

B. N2/v1: One-variable integral remainder

Object and role. Apply FTC and integrate by parts repeatedly.

Inputs. $g \in C^{(k+1)}$ near[0,1]; prior exports E1/v1

E — Ego

Own E2/v1: The exact one-variable Taylor formula with integral remainder.

M — Memory

Retain $g \in C^{(k+1)}$ near[0,1]; prior exports E1/v1; keep their assumptions and versions attached.

I — Intellect

Apply FTC and integrate by parts repeatedly.

S — Sense

The factorial and sign follow from the integration-by-parts recursion.

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For $k = 0$, $g(1) - g(0) = \int_0^1 g'(t)dt$ by FTC.

N2.2 For $k \geq 1$ integrate $I_k = (1/k!) \int_0^1 (1-t)^k \; g^{(k+1)}(t)dt$ by parts.

N2.3 The upper boundary is 0, the lower boundary contributes $-g^{(k)}(0)/k!$, and the remaining integral is I_{k-1}.

N2.4 Hence $I_k = I_{k-1} - g^{(k)}(0)/k!$.

N2.5 Iterating to $I_0 = g(1) - g(0)$ gives $g(1) = \sum_{j=0}^{k} g^{(j)}(0)/j! + I_k$.

Export E2/v1. The exact one-variable Taylor formula with integral remainder.

Acceptance gate. The factorial and sign follow from the integration-by-parts recursion. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: The factorial and sign follow from the integration-by-parts recursion. Recheck dependent nodes N3.

C. N3/v1: Restore tensors and bound the error

Object and role. Keep all k+1 factors of h in the remainder.

Inputs. Directional identities and scalar formula; prior exports E1/v1, E2/v1

E — Ego

Own E3/v1: Exact tensor expansion and a certified norm remainder bound.

M — Memory

Retain Directional identities and scalar formula; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

I — Intellect

Keep all k+1 factors of h in the remainder.

S — Sense

A scalar Hessian expression must not omit mixed terms or directional factors.

A — Action

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Substitute the node1 derivative identities into the formula to obtain exactly the asserted expansion for $f(a+h)$.

N3.2 If L bounds the operator norm of $D^{(k+1)}f$ on the segment, the remainder has modulus at most $L\|h\|^{(k+1)}/k!$ times $\int_0^1 (1-t)^k dt = L\|h\|^{(k+1)}/(k+1)!$.

N3.3 The bound is an additional reusable estimate with its explicit L; if $h = 0$ all positive-order terms and the remainder vanish.

Export E3/v1. Exact tensor expansion and a certified norm remainder bound.

Acceptance gate. A scalar Hessian expression must not omit mixed terms or directional factors. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: A scalar Hessian expression must not omit mixed terms or directional factors. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Restrict to a line

1. Define $g(t) = f(a + th)$.
2. Repeated chain rule along the affine path yields $g^{(j)}(t) = D^j f(a + th)[h, \dots, h]$ for $0 \le j \le k + 1$.
3. For $j = 0$ this means the function itself.
4. Compactness of the segment gives a neighborhood on which these restrictions are defined; no global convexity of U is assumed.

B. §4.N2 — One-variable integral remainder

1. For $k = 0$, $g(1) - g(0) = \int_0^1 g'(t)dt$ by FTC.
2. For $k \ge 1$ integrate $I_k = (1/k!) \int_0^1 (1 - t)^k \; g^{(k+1)}(t)dt$ by parts.
3. The upper boundary is 0, the lower boundary contributes $-g^{(k)}(0)/k!$, and the remaining integral is I_{k-1}.
4. Hence $I_k = I_{k-1} - g^{(k)}(0)/k!$.
5. Iterating to $I_0 = g(1) - g(0)$ gives $g(1) = \sum_{j=0}^{k} g^{(j)}(0)/j! + I_k$.

C. §4.N3 — Restore tensors and bound the error

1. Substitute the node1 derivative identities into the formula to obtain exactly the asserted expansion for $f(a + h)$.
2. If L bounds the operator norm of $D^{(k+1)}f$ on the segment, the remainder has modulus at most $L\|h\|^{(k+1)}/k!$ times $\int_0^1 (1 - t)^k dt = L\|h\|^{(k+1)}/(k + 1)!$.
3. The bound is an additional reusable estimate with its explicit L; if $h = 0$ all positive-order terms and the remainder vanish.

□

V. WORKED ORCHESTRATION

For $f(x,y) = e^{(x+y)}$ at $a = (0,0)$, $h = (u,v)$, $k = 1$, $Df(a)[h] = u + v$ and $D^2 f(th)[h,h] = e^(t(u+v))(u+v)^2$. Hence $e^{(u+v)} = 1 + u + v + (u+v)^2 \int_0^1 (1-t)e^(t(u+v))dt$. The remainder is at most $e^|u+v|(u+v)^2/2$.

A. Boundary and hypothesis audit

A first-order Taylor polynomial is generally not an exact value: $f(x) = x^2$ at 0 has zero first-order polynomial but remainder h^2. Sense requires the integral or a bound instead of dropping it. The theorem also cannot traverse a missing segment through a hole in U.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Ego → Memory:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Intellect → Action:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Sense → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Memory + Intellect:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Action:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Ego:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A first-order Taylor polynomial is generally not an exact value: $f(x) = x^2$ at 0 has zero first-order polynomial but remainder h^2. Sense requires the integral or a bound instead of dropping it. The theorem also cannot traverse a missing segment through a hole in U.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.